

Algorithms Study Guide

Coding Interview Prep — Sorting, Searching & Algorithmic Paradigms

1. Sorting & Searching Algorithms

Core techniques for organizing data and retrieving elements efficiently.

Algorithm	Time Complexity	Space Complexity	Key Characteristics
Bubble Sort	$O(n^2)$	$O(1)$	Simple comparison-based swap loop.
Merge Sort	$O(n \log n)$	$O(n)$	Divide and conquer; stable sorting.
Quick Sort	$O(n \log n)$ avg	$O(\log n)$	Pivot-based partitioning; in-place.
Linear Search	$O(n)$	$O(1)$	Sequential traversal; unsorted inputs.
Binary Search	$O(\log n)$	$O(1)$	Half-interval search; sorted inputs.

2. Algorithmic Design Paradigms

Foundational strategies for structuring solutions to complex engineering problems.

Paradigm	Core Concept	Practical Applications
Divide & Conquer	Breaks a problem into independent sub-problems, solves recursively, and combines results.	Merge Sort, Quick Sort, Binary Search.
Recursion	A function calling itself until reaching a terminal base case condition.	Tree traversals, call stack operations, backtracking.
Dynamic Programming	Solves overlapping sub-problems by storing sub-results (memoization/tabulation).	Fibonacci sequence, shortest paths, 0/1 Knapsack.
Greedy Algorithms	Makes locally optimal choices at each step to reach a global optimum.	Dijkstra's Algorithm, Huffman Coding, Fractional Knapsack.